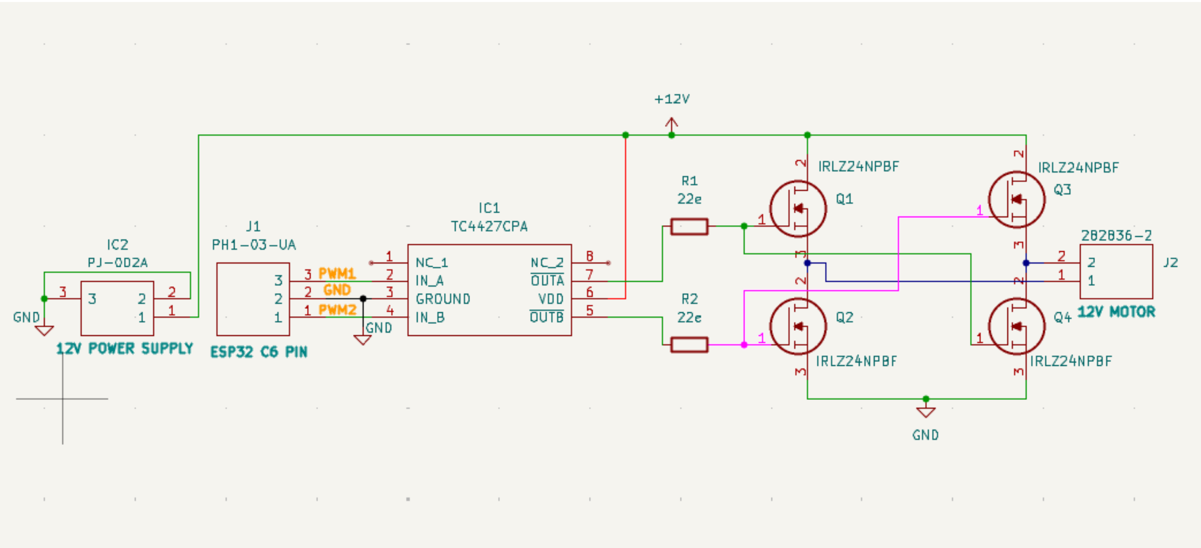
Dc Motor H-Bridge

A practical, step-by-step user manual that walks you from a blank project to production-ready Gerbers using KiCad. Intended for beginners and intermediate users who want a clean, repeatable PCB design workflow.

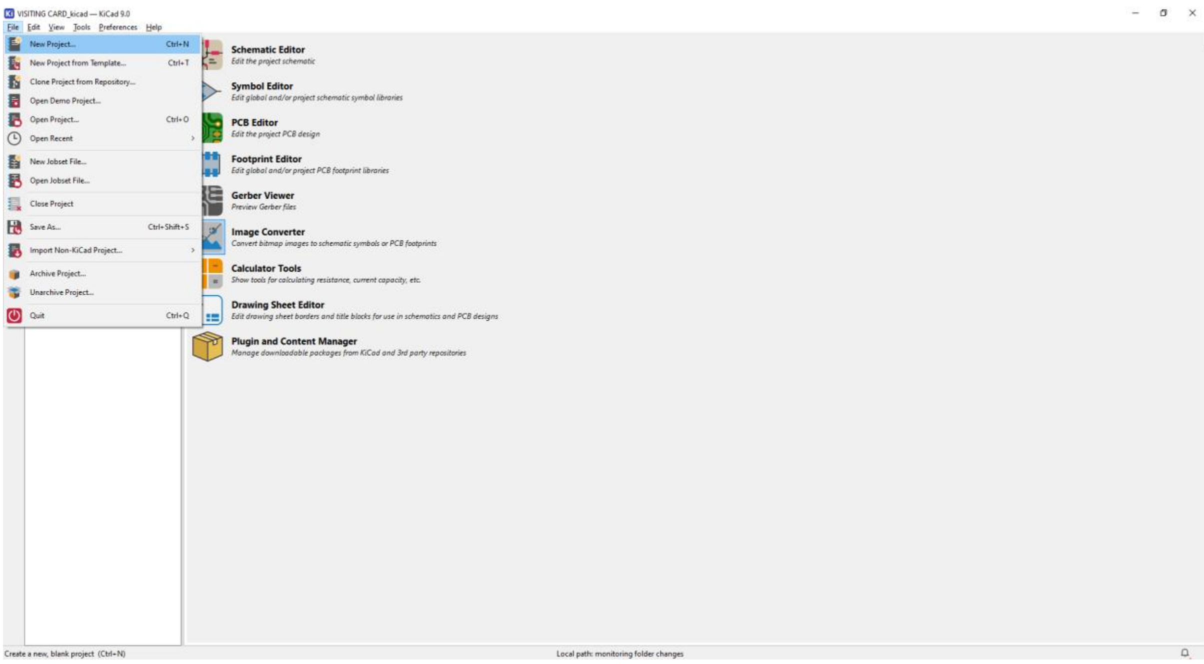
Circuit diagram:



Start a project:

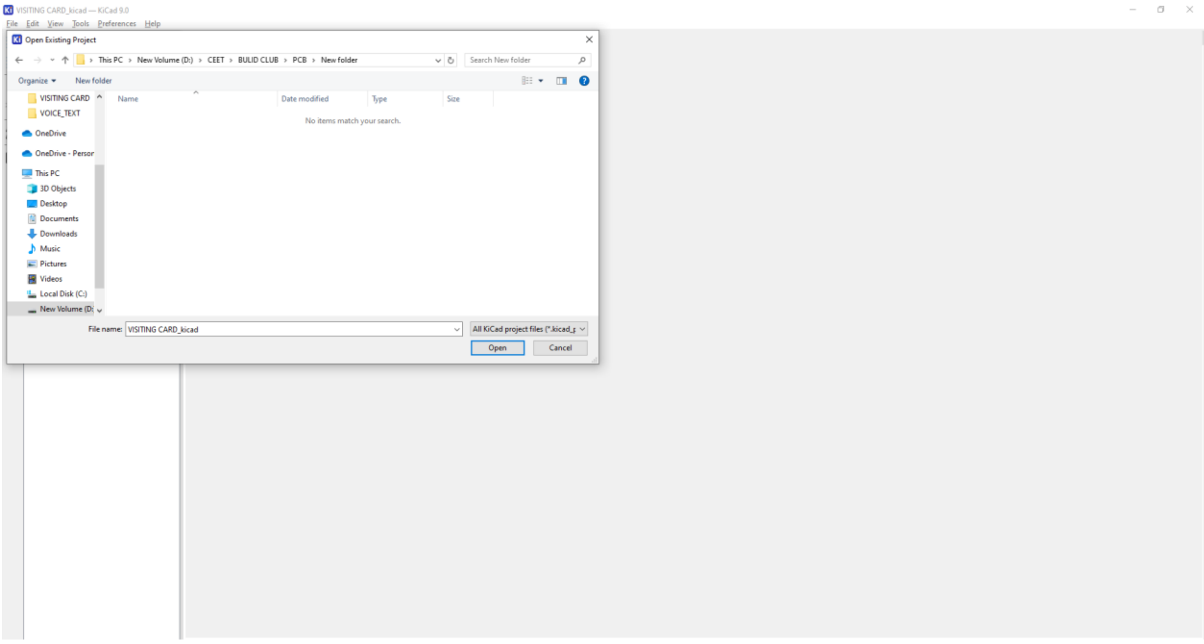
Step1:

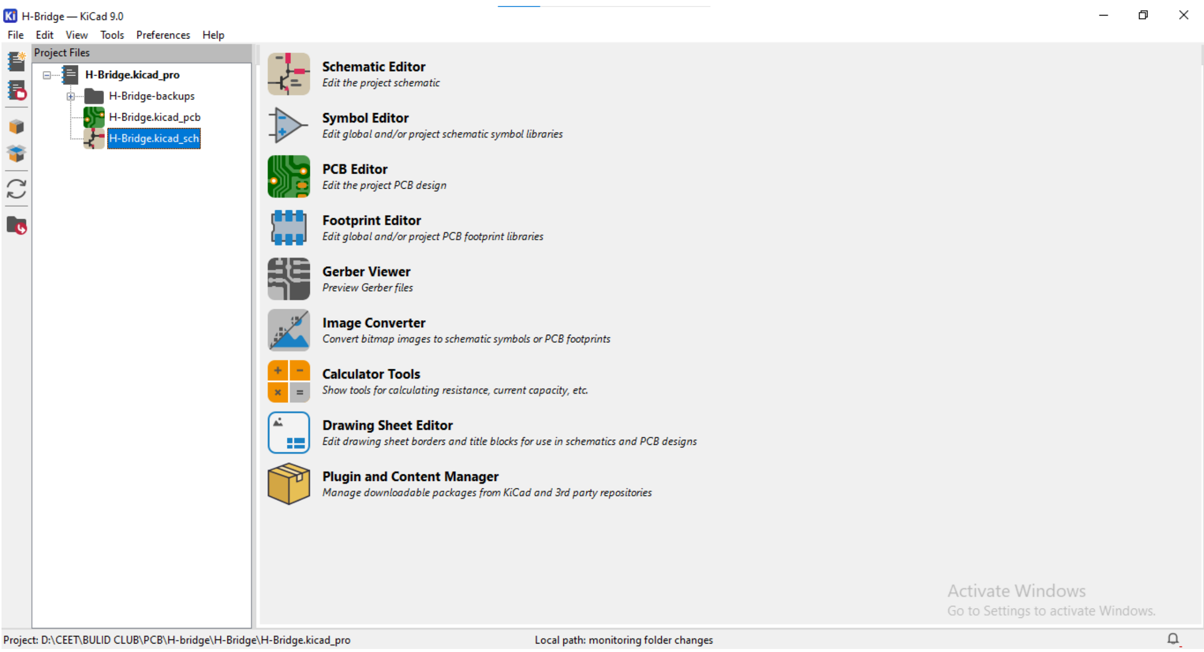
- File → New Project → give a clear name



Step2:

- Enter the project title
- Save in a folder structure.





Step 3:

- Circuit components :

S. No	Components	Quantity
1.	IRLZ24NPBF Mosfet	4
2.	TC4427 Gate Drivers	1
3.	22 ohms (Through hole) Resistor	2

4.	PJ-002A Power Connector	1
5.	PH1-03-UA Berg strip male connector	1
6.	282836-2 motor connector	1

Some components are not there in the kicad Ide so externally download the components after import the kicad ide.

- IRLZ24NPBF Mosfet download- link
- TC4427 Gate Drivers download- link
- PJ-002A Power Connector download- link
- PH1-03-UA Berg strip male connector download- link
- 282836-2 motor connector download- link

Go to download the All components.

Step 4:

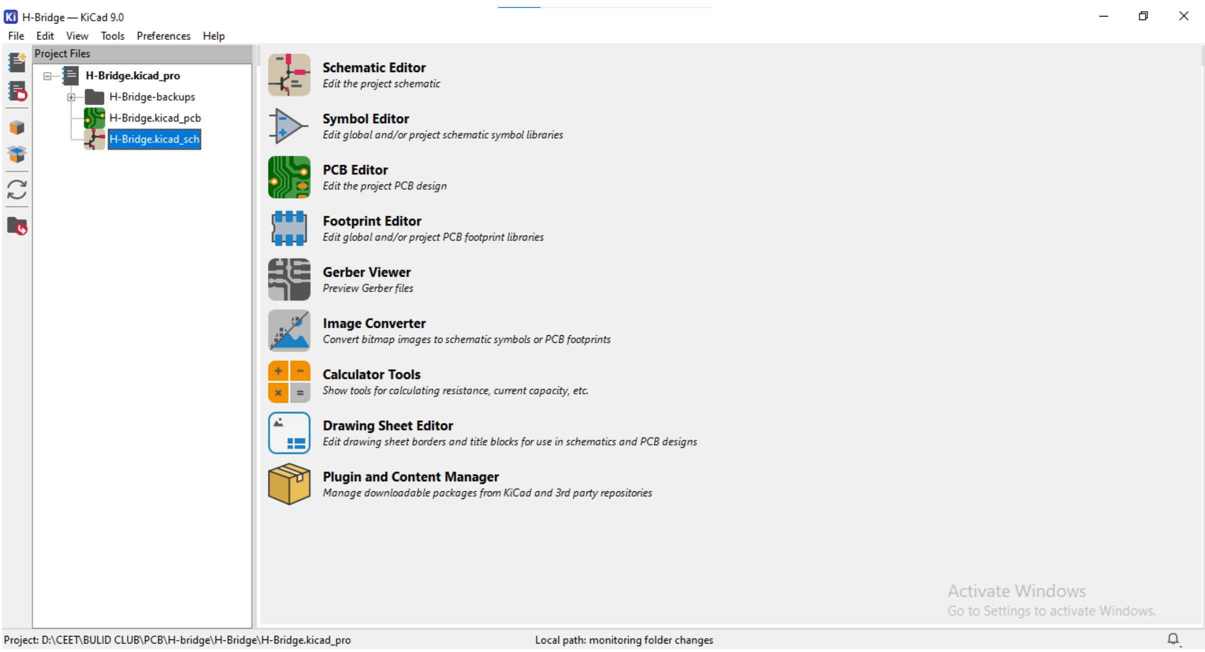
- Download the components after extracting the ZIP file.
- Go to Preferences > Manage Symbol Libraries Open it
- Add existing library to table option has been open it.
- Open the components folder, select the kicad folder
- Select the Kicad Library file
- After Symbol Libraries window has ok option click it.

Step 5:

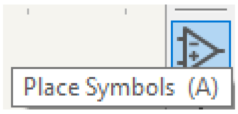
- Go to Preferences > Manage Footprint Libraries Open it
- Add existing library to table option has been open it.
- Select the legacy
- Open the components folder, select the kicad folder.
- Select the Kicad Library file
- After Footprint Libraries window has ok option click it.

Step 6:

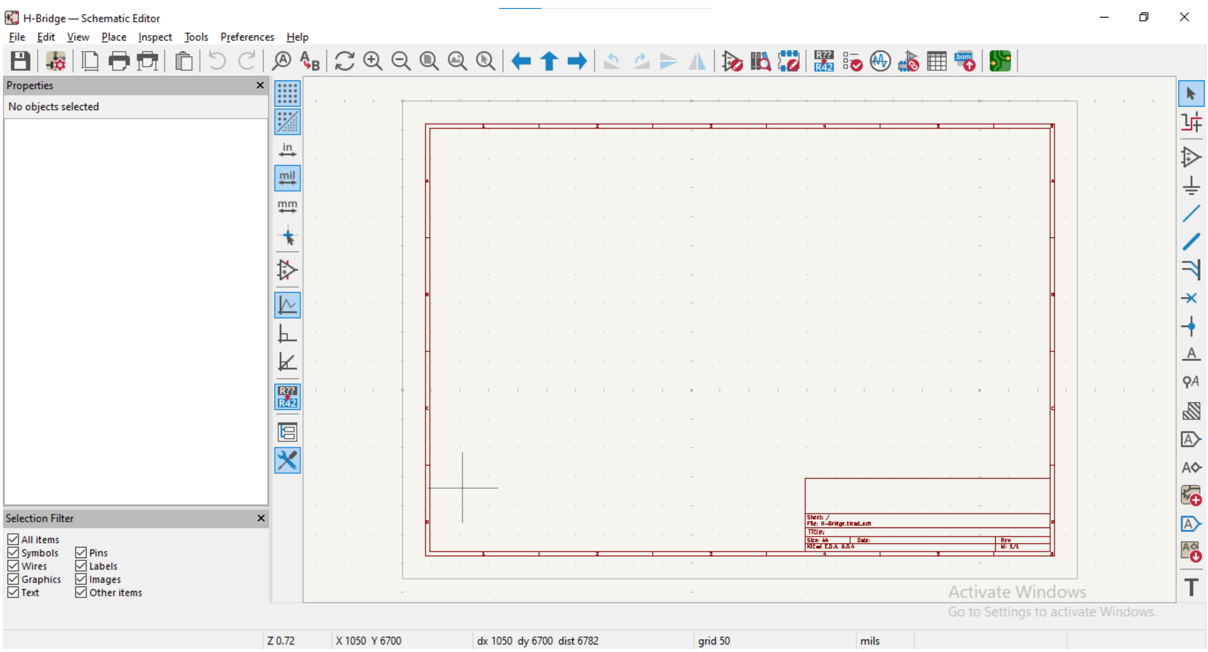
- Open the schematic editor file



- Click the Place Symbols ion,



- On your right-side top corner / press ‘A’ key word

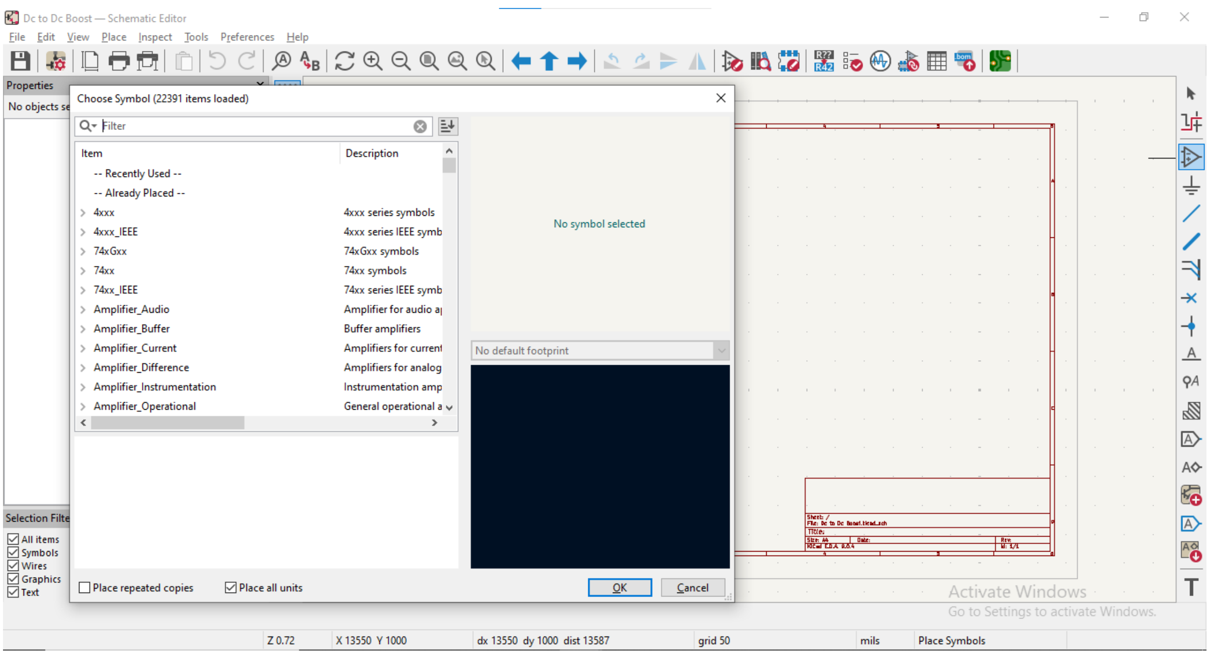


Step 7:

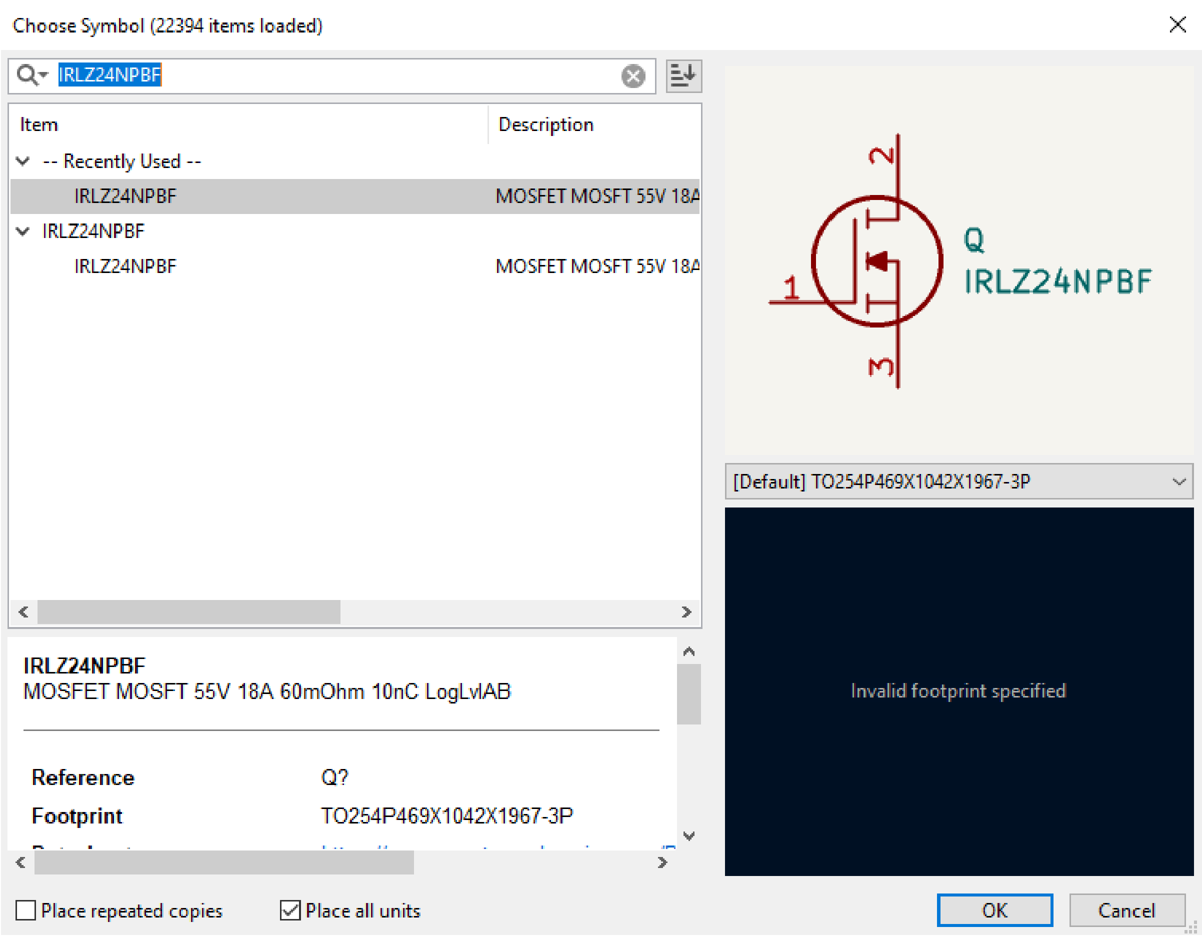
- Select your circuit components

S. No	Components	Quantity
1.	IRLZ24NPBF Mosfet	4
2.	TC4427 Gate Drivers	1
3.	22 ohms (Through hole) Resistor	2
4.	PJ-002A Power Connector	1
5.	PH1-03-UA Berg strip male connector	1
6.	282836-2 motor connector	1

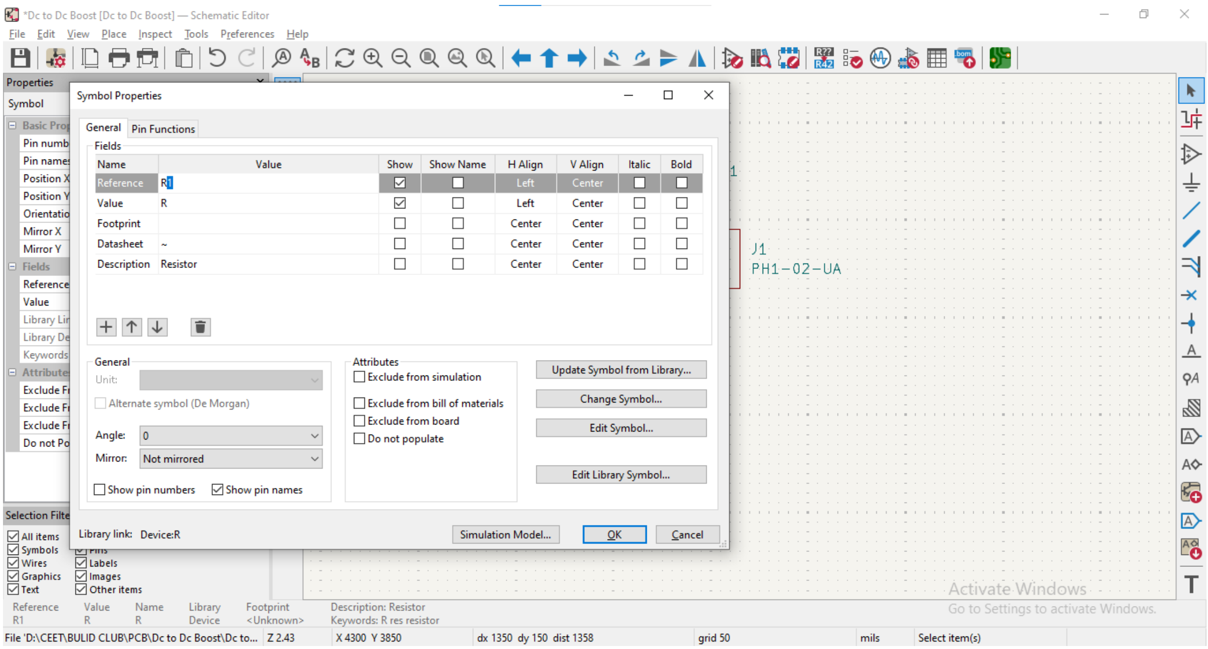
- Go to filter option
- Seach the components name or part number (ex-inductor or IRLZ24NPBF)



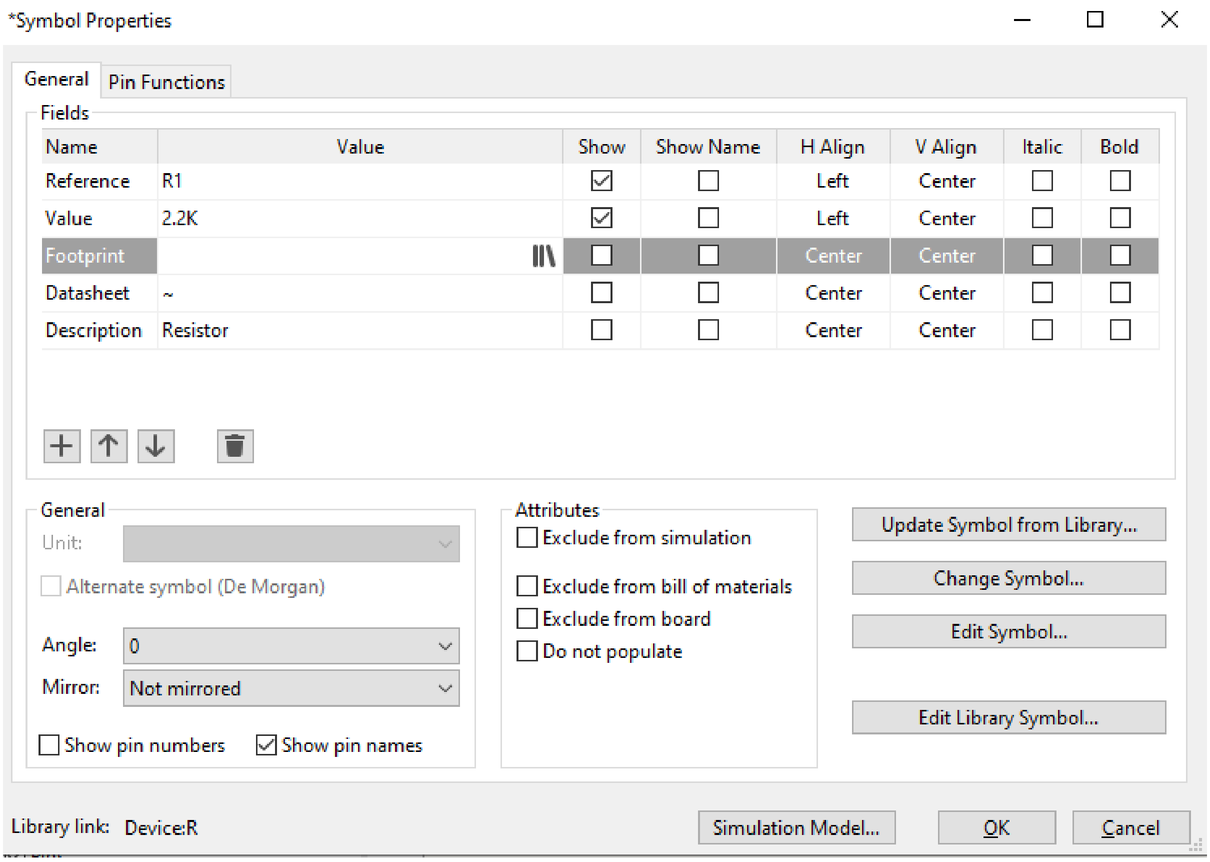
- Select your component after placing it



- To place all components on the sheet
- Select the component press ‘E’ add the value (Resistor, inductor) and all components add the footprint

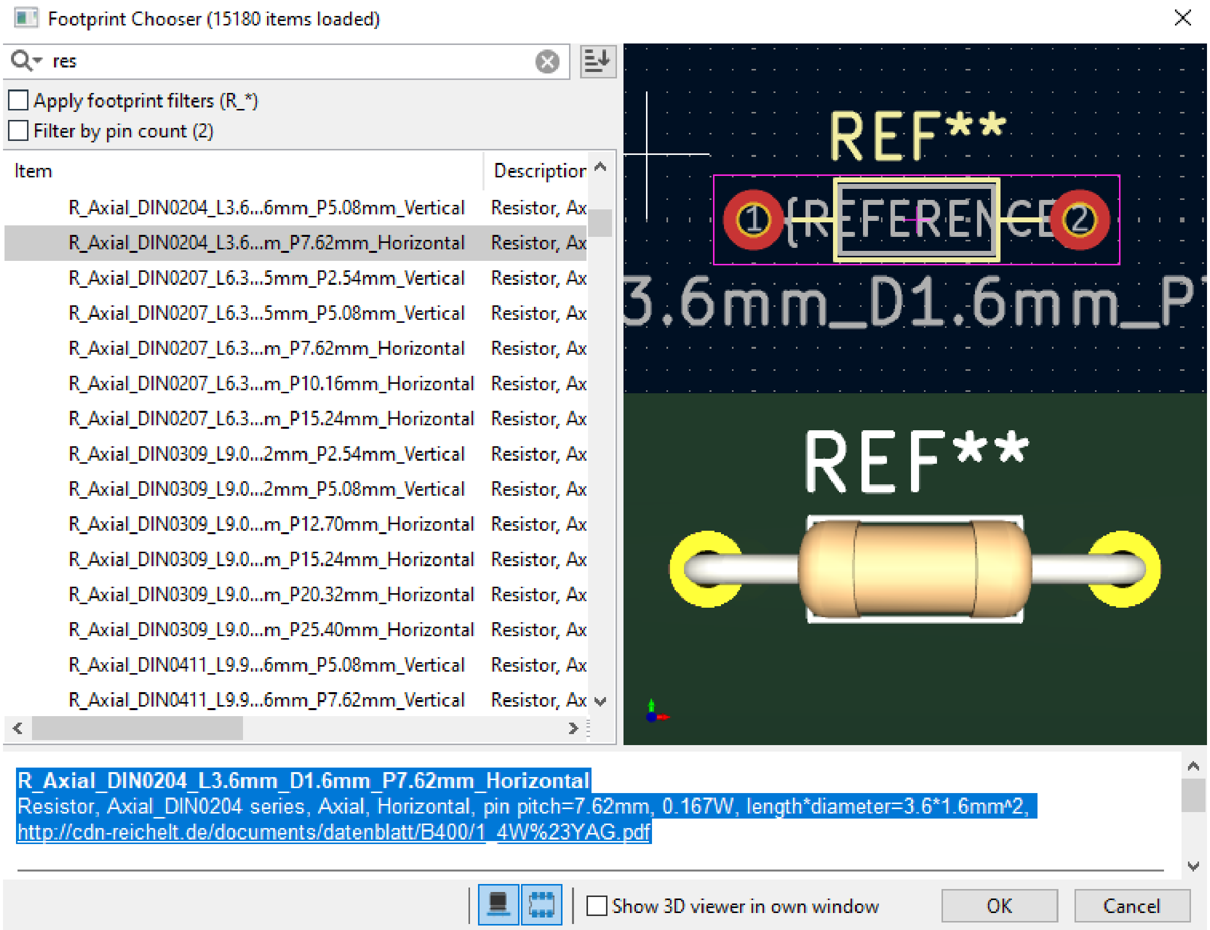


- Give the value like that (2.2k ohms or 100uf)

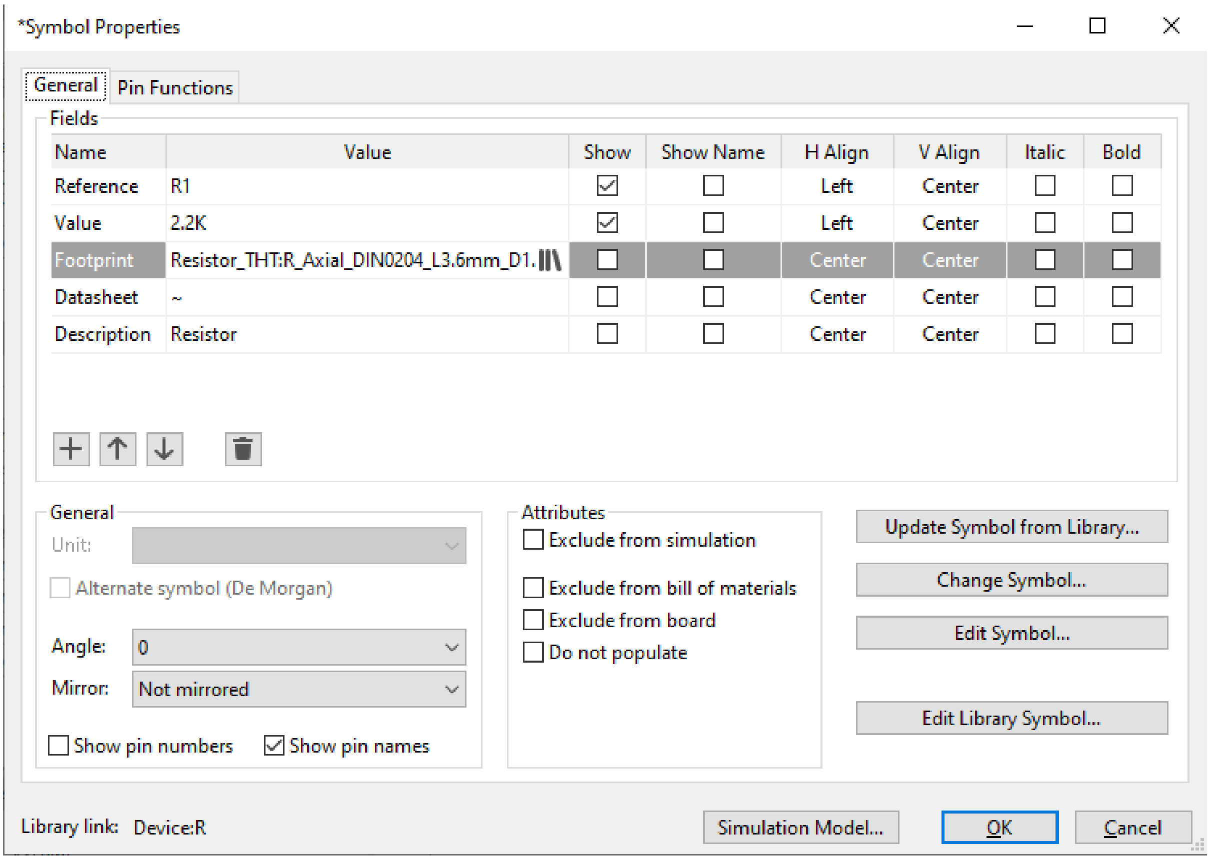


- Press the footprint option
- Select the part number (like a package-through hole)
- Go to filter option

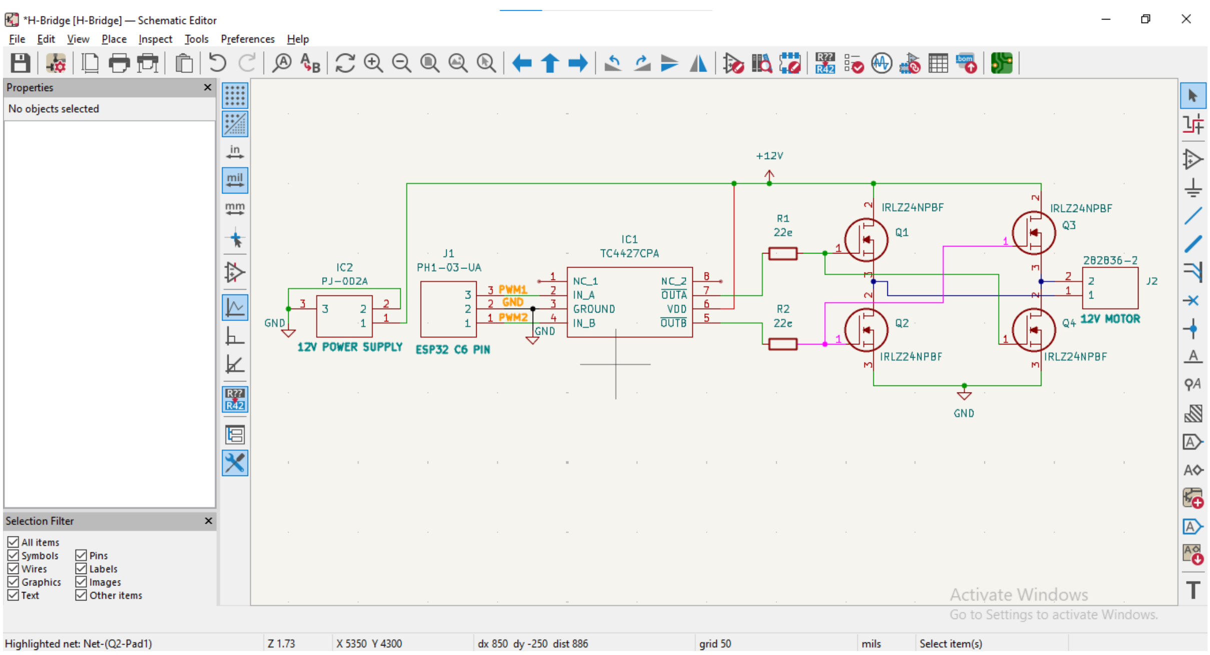
- Seach the components name or part number



- Select your component Footprint after ok it
- Symbol Properties finally ok option clicks



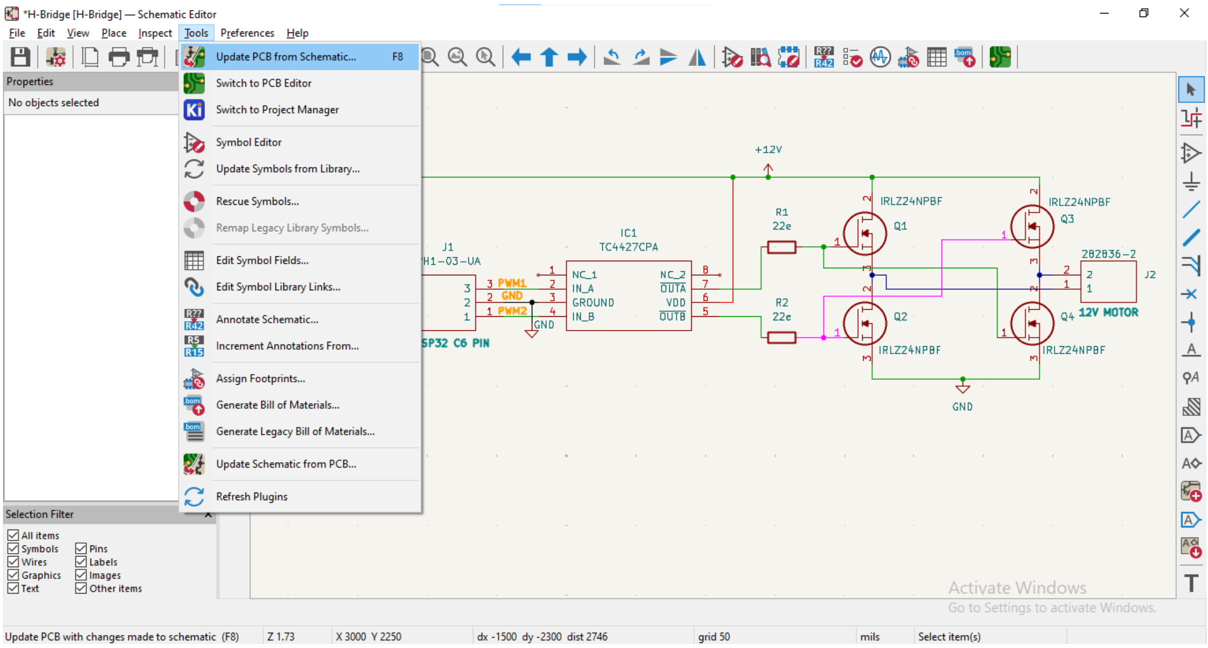
- Next give all components value has enter symbol properties
- Connect symbols with wires
- Press W key word



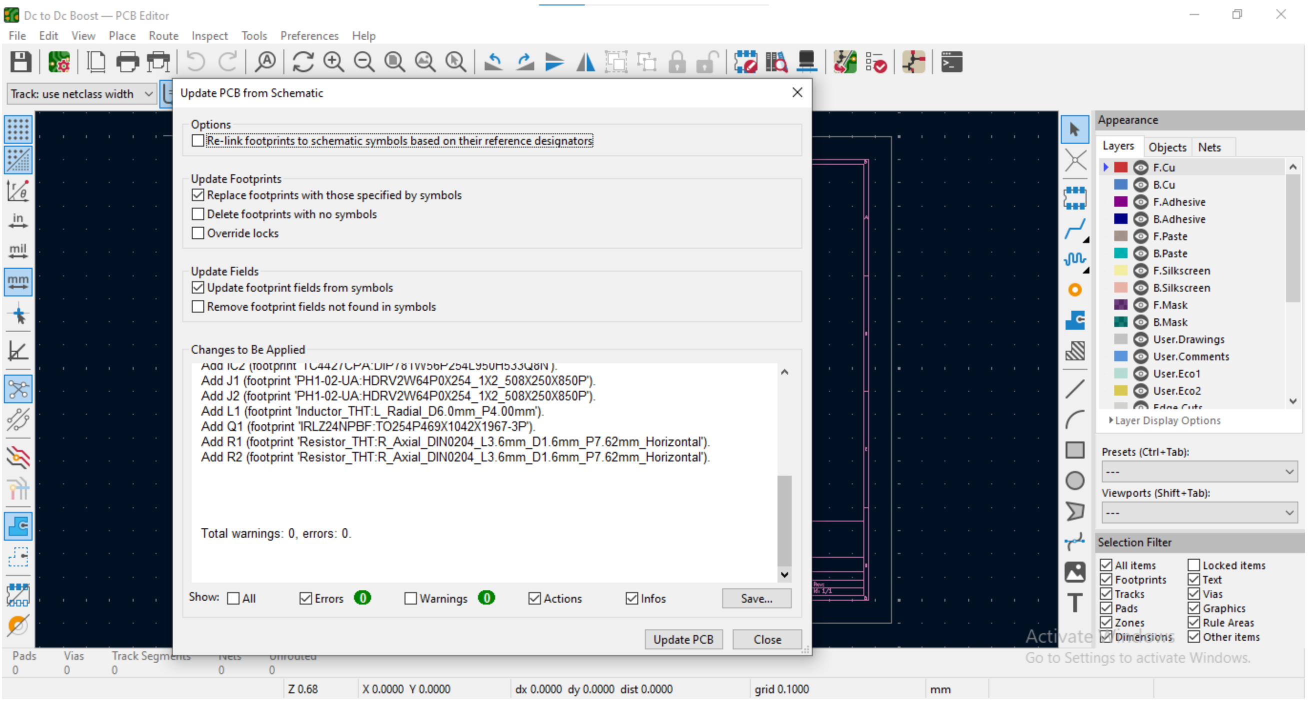
- Save the Schematic file.

Step 8:

- Go to ‘Tool’ → ‘Update PCB from schematic’ / F8



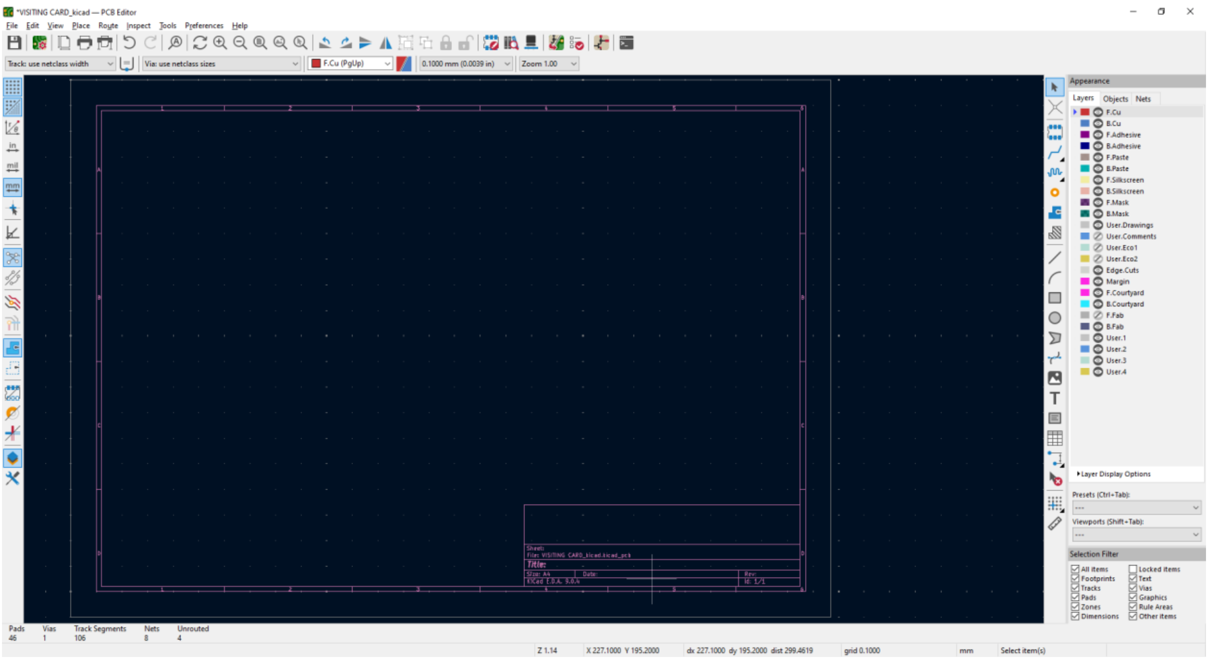
- Check each schematic symbol must have a matching PCB footprint
- If any footprint has missing it can’t update PCB, it’s shown error warning



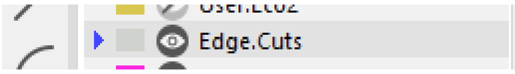
- Go to clear the error (like a No unconnected, annotate symbols, Fix wiring errors, missing power flags) Otherwise update the PCB option click

Step 9:

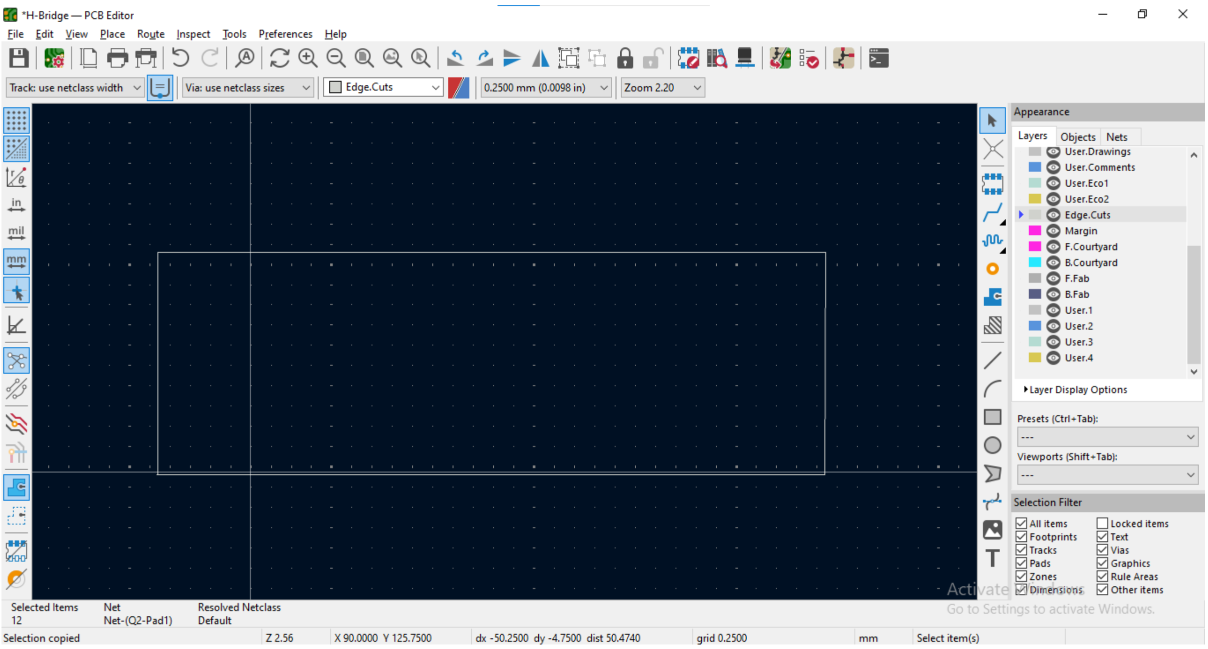
- Set the board outline (Edge Cuts layer)



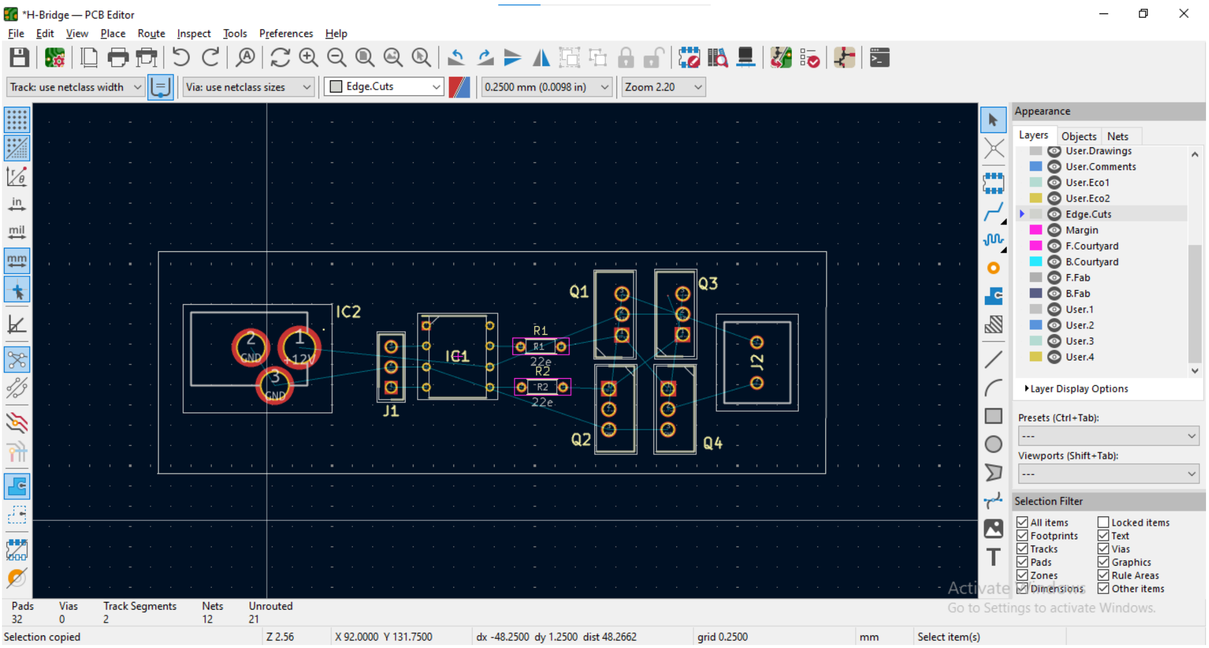
- Click the Place Symbols ion, On your right-side corner



- Select the rectangle tool using to set the board outline

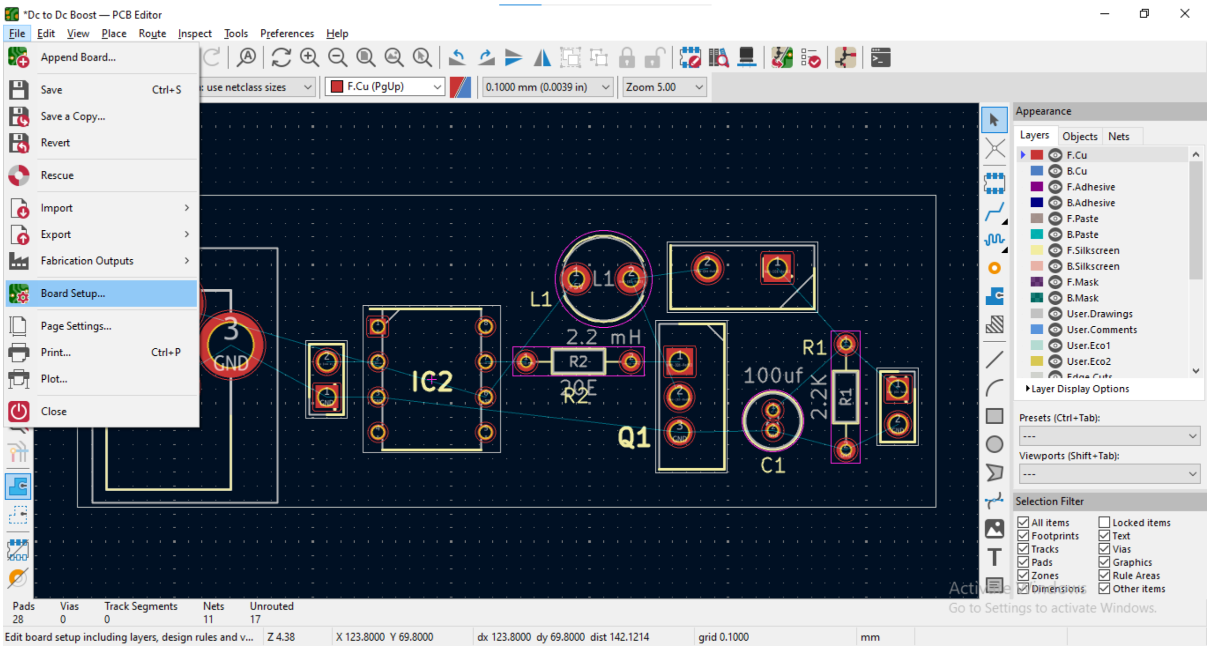


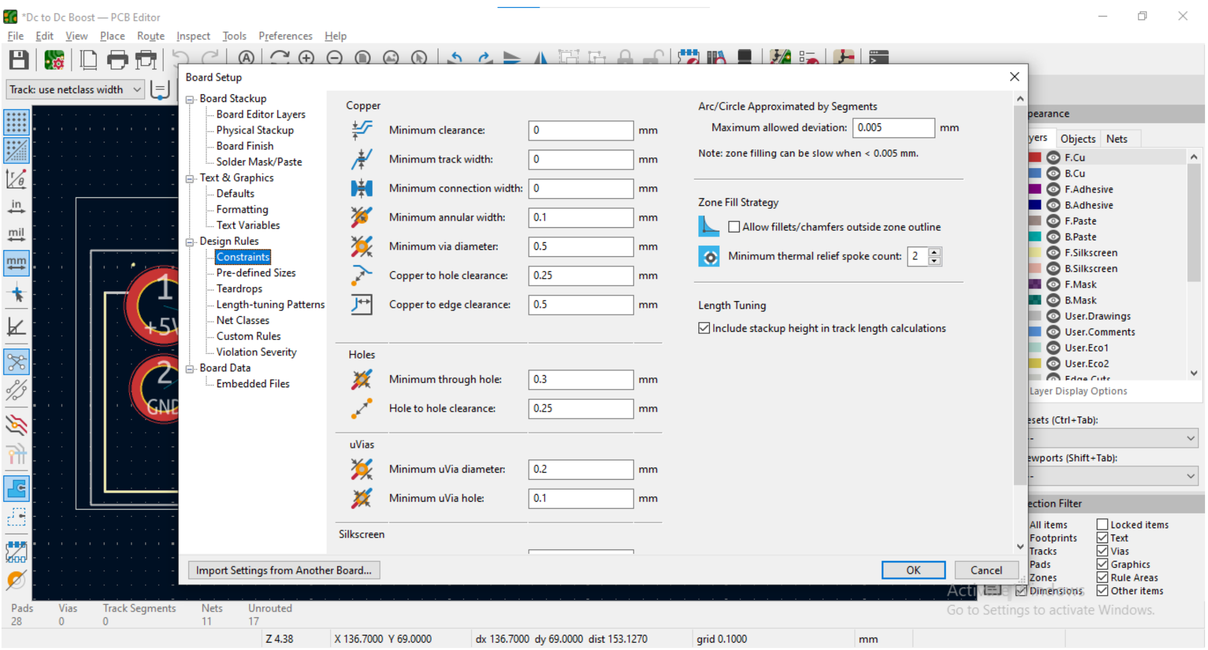
- To place all components on the sheet



Step 10:

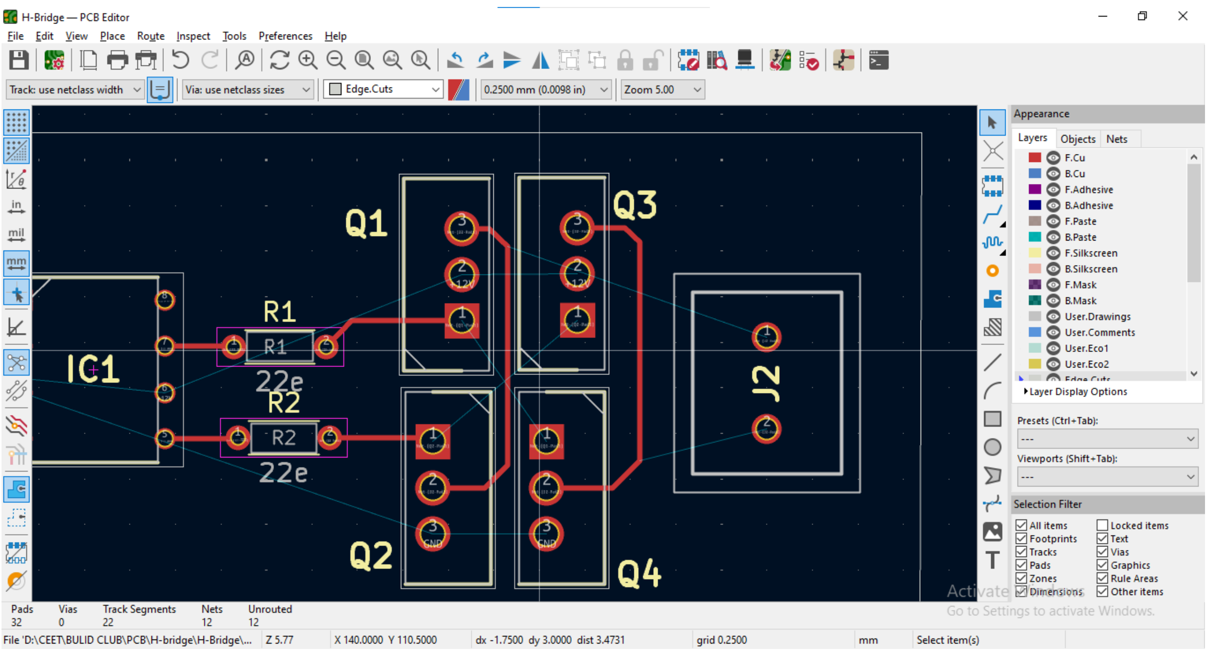
- Next go to File → Board Setup → Constraints
- Set the details of all option
- And refer the link (**Disclaimer** – 'Constraints' depends on PCB manufacturer)



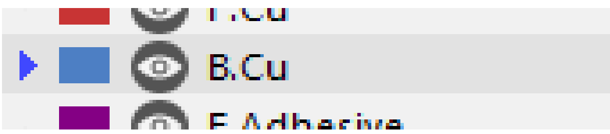


Step 11:

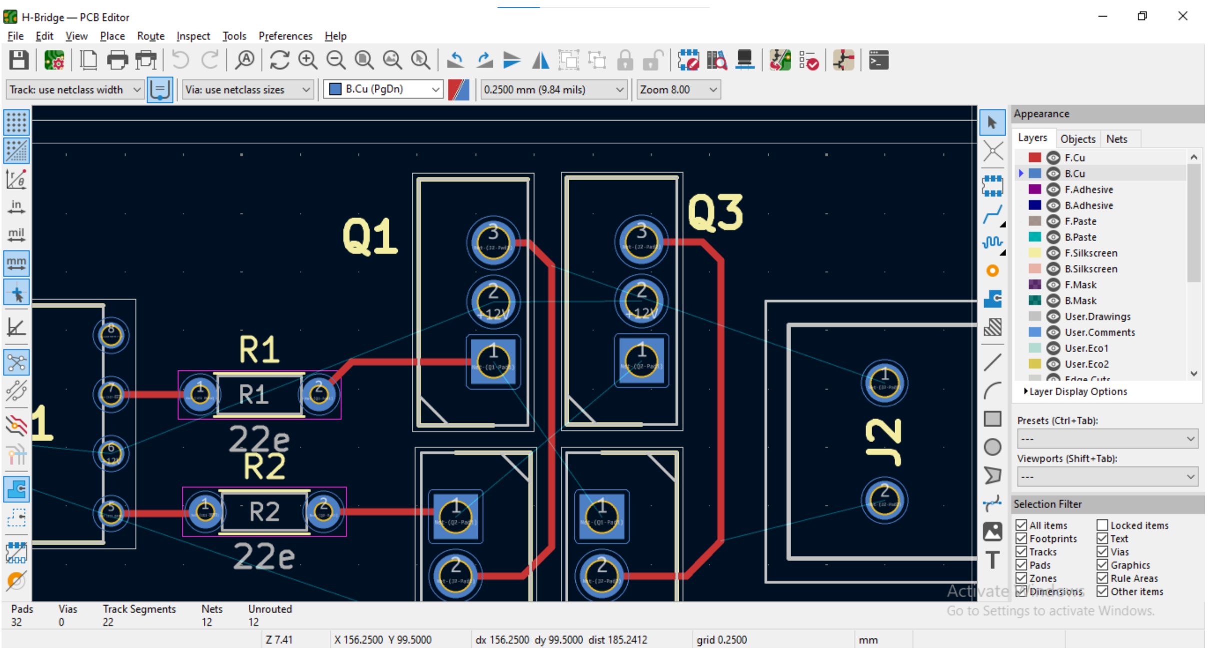
- Go to select the layer after routing in the traces short key "X"
- Traces thicker increase or decrease width via net class selection based



- Go to select the bottom layer
- Click the Place Symbols ion, On your right-side corner



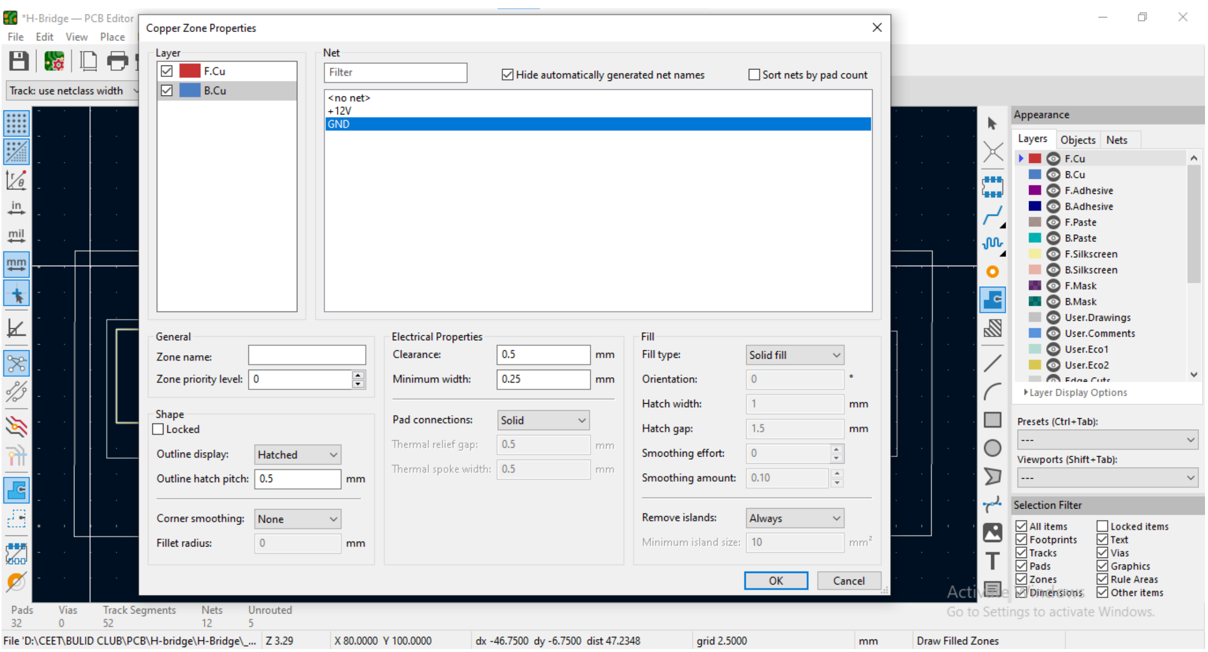
- Go to select the layer after routing in the traces short key "X"



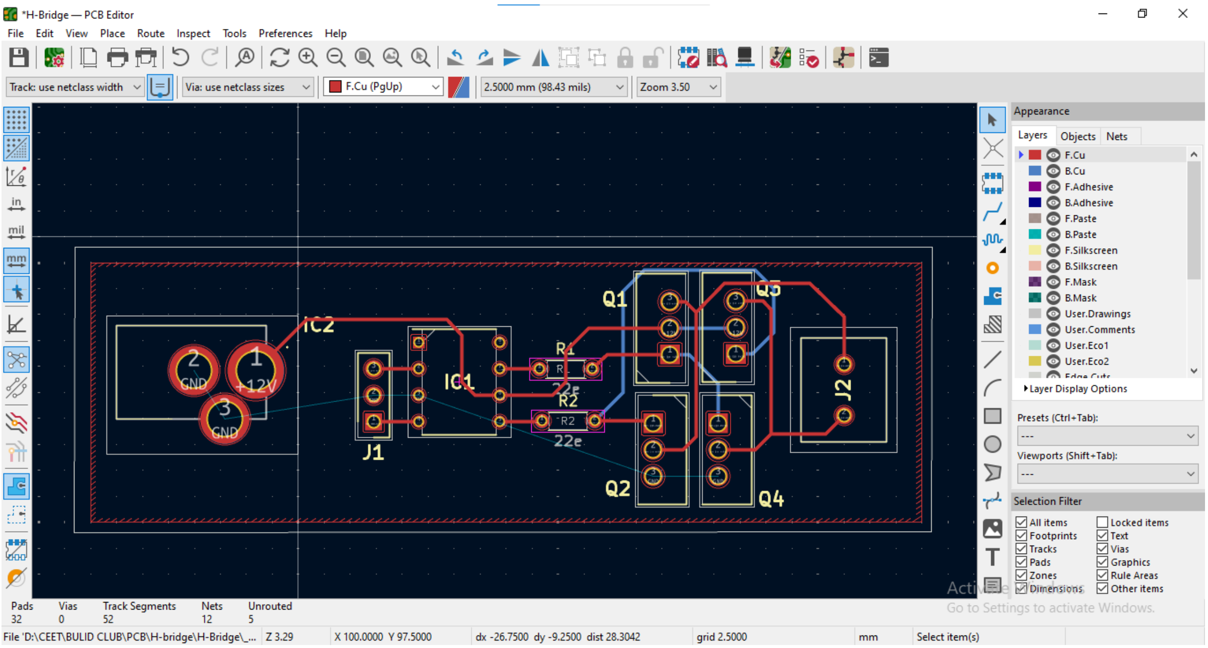
- Click the Place Symbols ion, after click the PCB grid

 Polygon tool

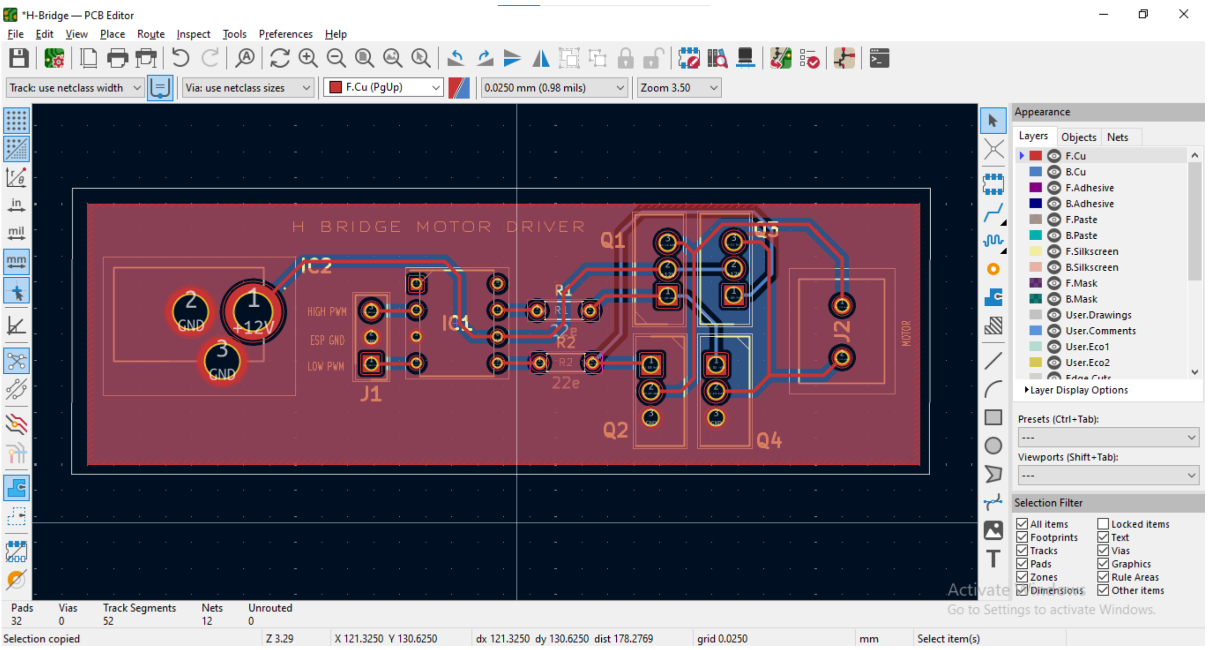
- It's automatically open Copper Zone Properties, selected the GND net



- Polygon using to draw the rectangle

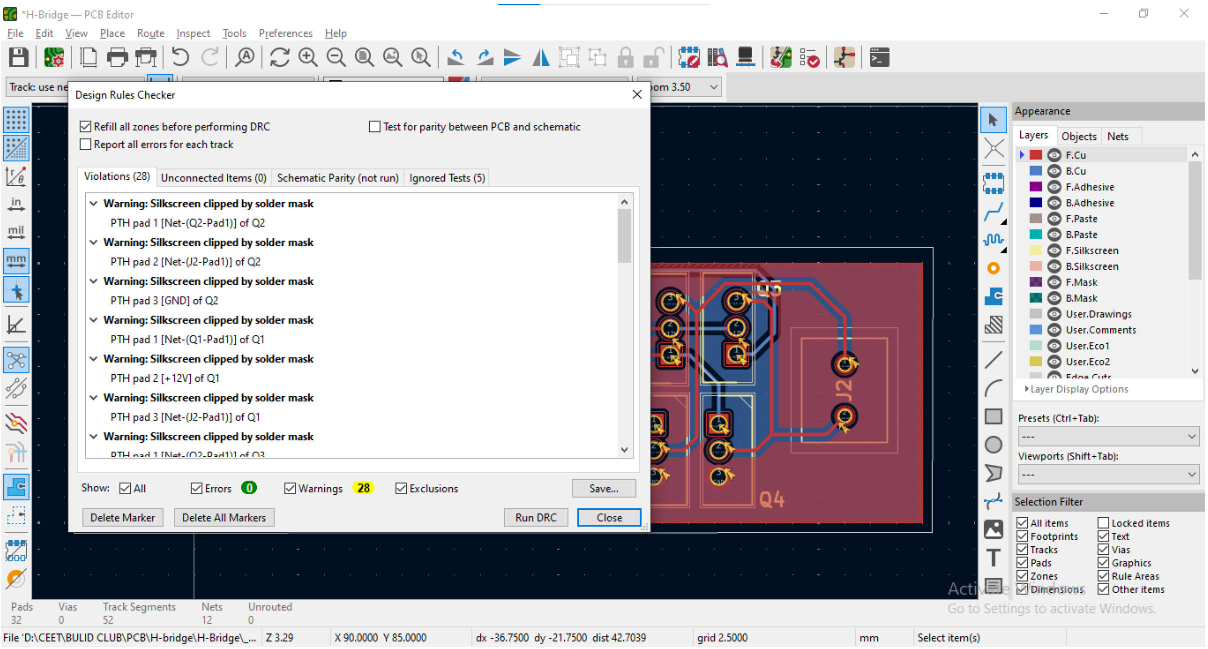


- Press the key “b” or go to edit option → Fill All Zones click it



Step 12:

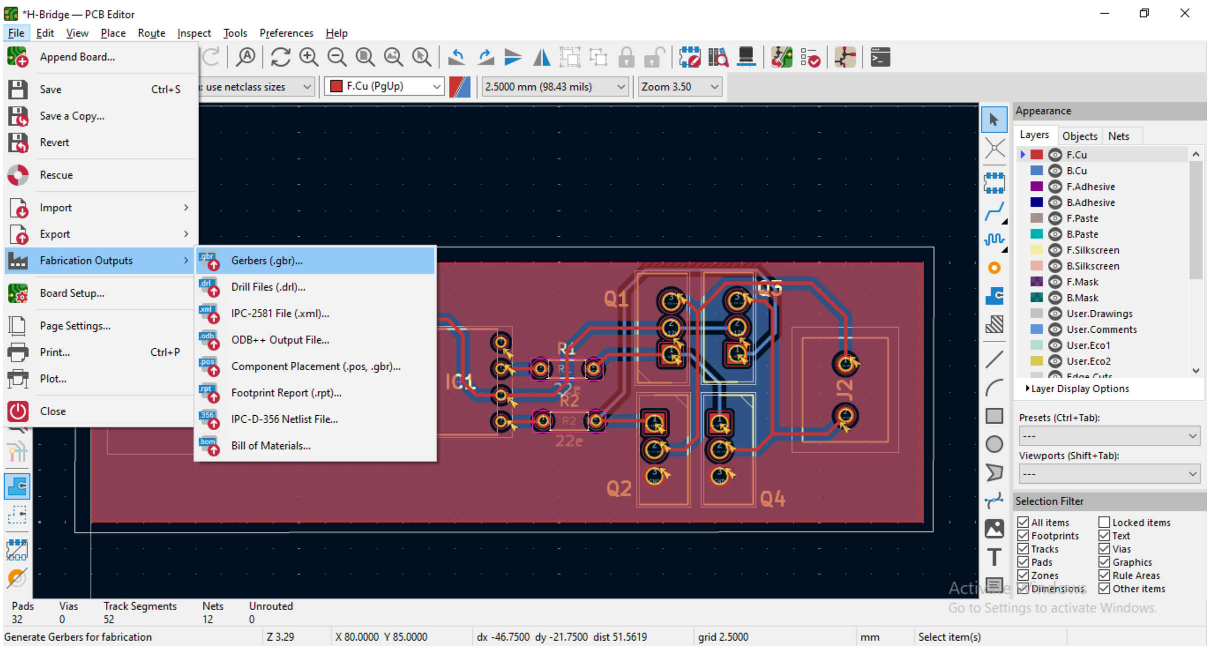
- Go to ‘Inspect’ → ‘Design Rules checker’
- Check each Constraints value must have a matching PCB design
- If any Constraints value has missing it can’t run DRC, it’s shown error warning Go to clear the error (clearances, unconnected pins, minimum annular rings)



- If clear all error go to Fabrication outputs

Step 13:

- Go to file →Fabrication outputs → Gerber, click it

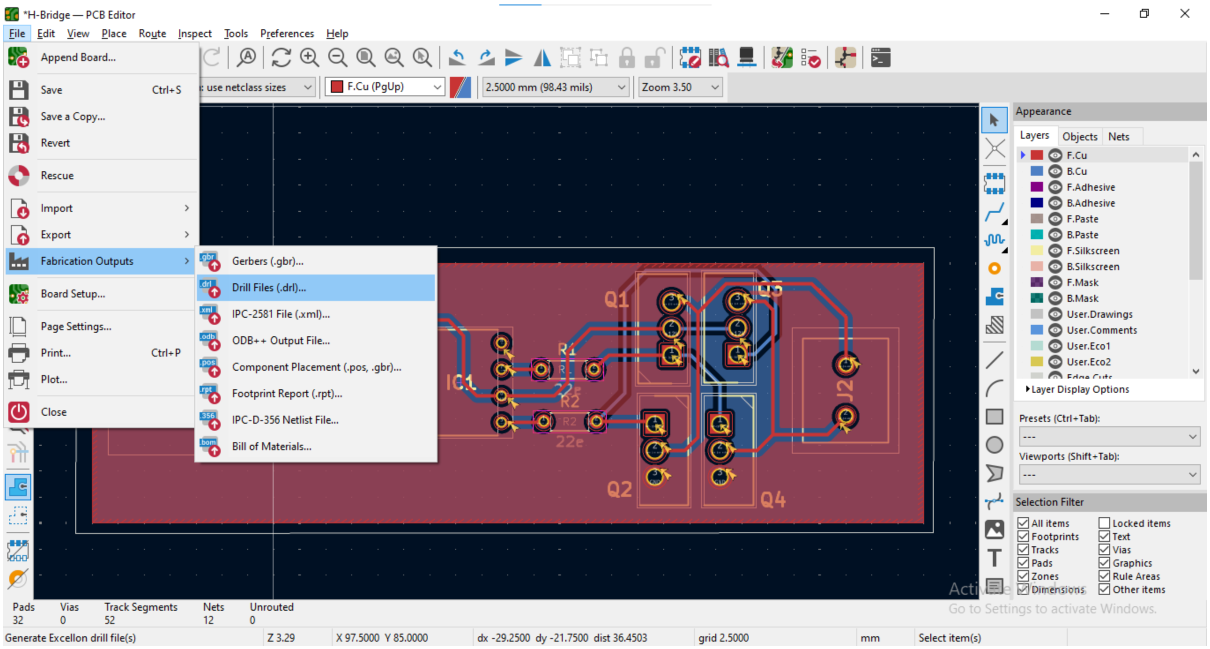


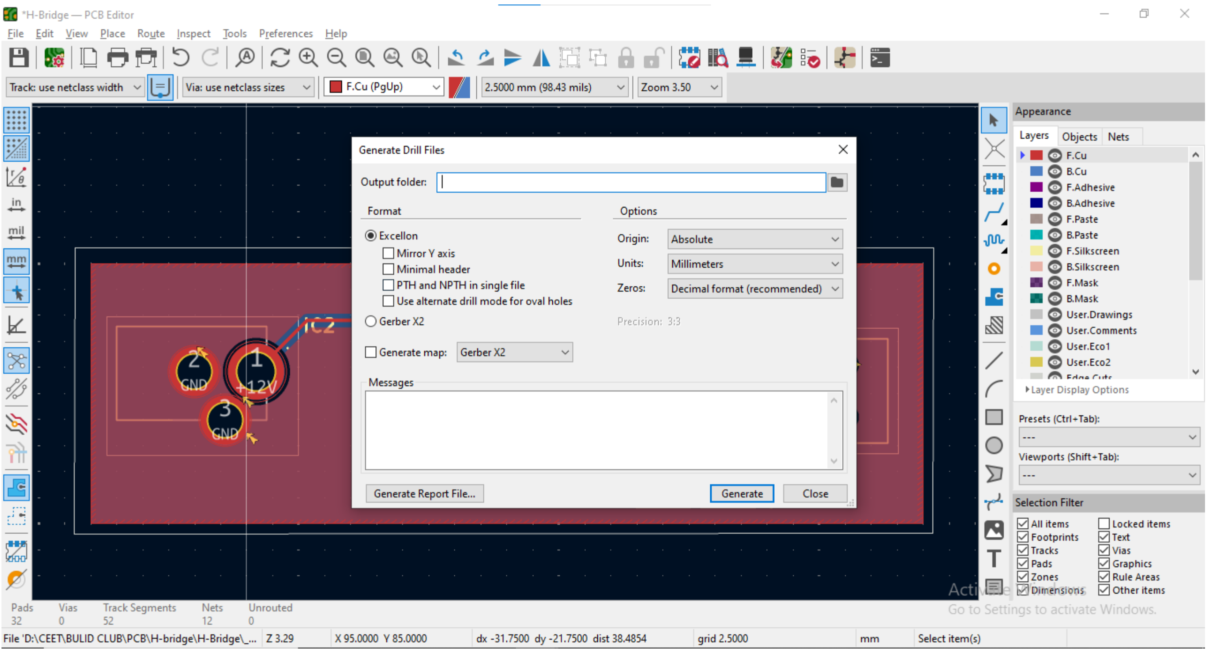
- Select all layers

  — F. Cu (Top copper)
  — B. Cu (Bottom copper)
  — F. Silk (Top silkscreen)
  — B. Silk (Bottom silkscreen)

— F. Mask (Top solder mask)
— B. Mask (Bottom mask)
— Edge. Cuts (Board outline)
— Drill file (Excellon)

- Next Click the Generate Drill Files option





- Finally, Dc12v motor driver PCB is ready
- Please find the reference video link for creating a schematic in KiCad IDE link
- Reference link